

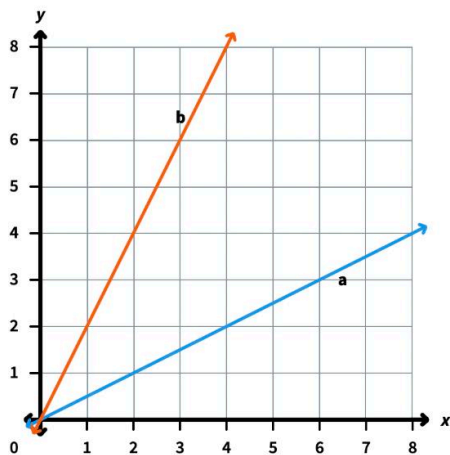
Mini Lesson Sample Script

Materials

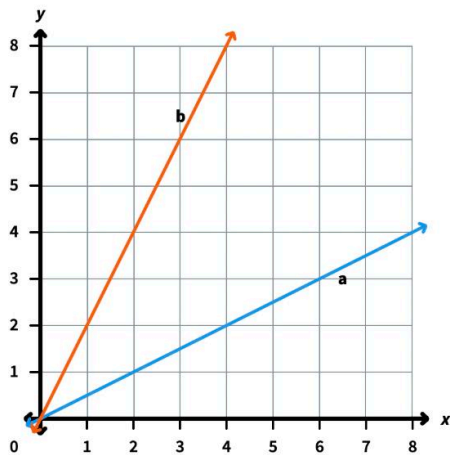
(S) Mini-Lesson Worksheet, personal white board

PROBLEM

1 The table below shows Biscuit the dog's weight at different ages.



Plot each data point on the coordinate plane below. Then, connect the points with line segments to make a line graph.



Answer: Points plotted at (0, 3), (4, 15), (8, 25), (12, 35), (16, 40), (20, 45), (24, 45), (28, 45) and connected with line segments.

SAMPLE SCRIPT

T: Here's a table showing how much a dog named Biscuit weighed at different ages. We're going to use this data to build a line graph. Take a look at the coordinate plane. What does the x-axis show?

S: Biscuit's age in months.

T: And the y-axis?

S: His weight in pounds.

T: Right. Let's plot the first point together. At 0 months, Biscuit weighed 3 pounds. So I start at 0 on the x-axis and go up to 3 pounds on the y-axis. That's the point (0, 3). Plot it on your graph.

S: (Plot.)

T: At 4 months, Biscuit weighed 15 pounds. I move to 4 on the x-axis and go up to 15 pounds. Plot that one too.

S: (Plot.)

T: Good. Now use the table to plot the rest of the points on your own.

S: (Plot remaining points.)

T: Now connect all the points with line segments, starting from the first point to the second, the second to the third, and so on.

S: (Connect the points.)

T: Now we have a line graph that shows how Biscuit's weight changed over time.

2 Use the line graph to find how much weight was gained during different time periods by subtracting coordinates.

Use the line graph of Biscuit's weight to answer the following questions.

- a. How much weight did Biscuit gain in the first 4 months of his life, between 0 months and 4 months?
- b. How much weight did Biscuit gain in the second 4 months of his life, between 4 months and 8 months?
- c. How much more weight did Biscuit gain in the first 4 months than in the second 4 months?

Answer: a) 12 pounds; b) 10 pounds; c) 2 pounds

T: Look at the line graph. What did Biscuit weigh at 0 months?

S: 3 pounds.

T: And at 4 months?

S: 15 pounds.

T: So how can we figure out how much weight he gained in those first 4 months?

S: Subtract. 15 pounds minus 3 pounds is 12 pounds.

T: So Biscuit gained 12 pounds in the first 4 months. The difference in the y-coordinates tells us how much weight he gained. Now try part (b). How much weight did Biscuit gain between 4 months and 8 months?

S: At 4 months he weighed 15 pounds, and at 8 months he weighed 25 pounds. 25 pounds minus 15 pounds is 10 pounds.

T: So he gained 10 pounds. Now for part (c) — how much more weight did Biscuit gain in the first 4 months than in the second 4 months?

S: 12 pounds minus 10 pounds is 2 pounds. He gained 2 more pounds in the first 4 months.

3 Analyze the line graph to identify when growth was fastest, slowest, or unchanged by comparing the steepness of line segments.

Use the line graph of Biscuit's weight to answer the following questions.

- a. During which 4-month period did Biscuit gain weight the fastest? How can you tell from the graph?
- b. What happened to Biscuit's weight between the ages of 20 months and 28 months? How does the line graph show this?

Answer: a) Biscuit gained weight the fastest between 0 months and 4 months because that line segment is the steepest; b) Biscuit's weight

T: Look across the whole line graph and answer parts (a) and (b) on your own.

S: (Work.)